

Rank Before You Merge: Training-Free, Merger-Aware Visual Token Compression for Vision-Language Models

Zhengxing Yan¹, Hao Sun², Chen Guo³, and Jianpeng Hu^{1,*}

¹Shanghai University of Engineering Science

²Nanjing University of Science and Technology

³East China Normal University

super_fw@163.com; sh1076849939@163.com

guochenjkhjk@qq.com; mr@sues.edu.cn

*Corresponding author

Abstract

Most VLM token selectors rank outputs of a native spatial merger, implicitly assuming that merging preserves which evidence is salient. We show that a learned merger can instead rewrite feature-norm rankings and demote units that contain text strokes. Rank-Before-Merge (RBM) addresses this operator-order failure by ranking complete native units before the original merger, under a fixed per-image budget and unchanged language-model interface. At 25% retention, RBM outperforms matched post-merger L2 selection in all nine text-dense model–benchmark pairs across Qwen3-VL, Qwen2.5-VL, and InternVL3, by 11.0–38.4 percentage points and 235–432 OCRBench points. Rank, edge-energy, and swap controls trace the gap to merger-induced saliency rewriting. A pre-merger prior also improves FastV-k3 on four locked Qwen3-VL probes, whereas the GQA reversal marks the workload boundary. RBM is training-free, adds no parameters, and raises measured Qwen3-VL throughput by 68% at 25% retention. Code: github.com/Code-Yan-ZX/qrbm-vlm.

Introduction

Modern VLMs expose hundreds to tens of thousands of visual tokens to the language model. Retaining 25% of Qwen3-VL’s visual tokens lowers latency on our A40 from 0.36 to 0.23 seconds and raises throughput by 68%. Compression must therefore decide which evidence should occupy the limited rate.

Most selectors rank tokens after a native spatial merger combines neighboring patches. This order assumes that merging preserves the evidence that would have ranked highest before merging. A learned merger can instead reverse the top- k order, hiding answer-bearing regions before downstream selection begins. The representation on which top- k acts thus changes task distortion even at the same rate. We ask whether ranking before the merger reduces this distortion when the budget, merge units, and downstream model are fixed.

Rank-Before-Merge (RBM) makes one intervention: it scores complete 2×2 units at the merger input, then sends an exact per-image budget through the unmodified merger and language-model interface (Fig. 1). Its units, positions, stream identity, and interface follow the deployed graph; we call this contract *machine-centric*.

At 25% retention, RBM improves all nine text-dense full-split comparisons across three VLMs and two mergers; GQA favors post-merger selection. Rank and token-survival analyses show that the merger rewrites the selected set and disproportionately

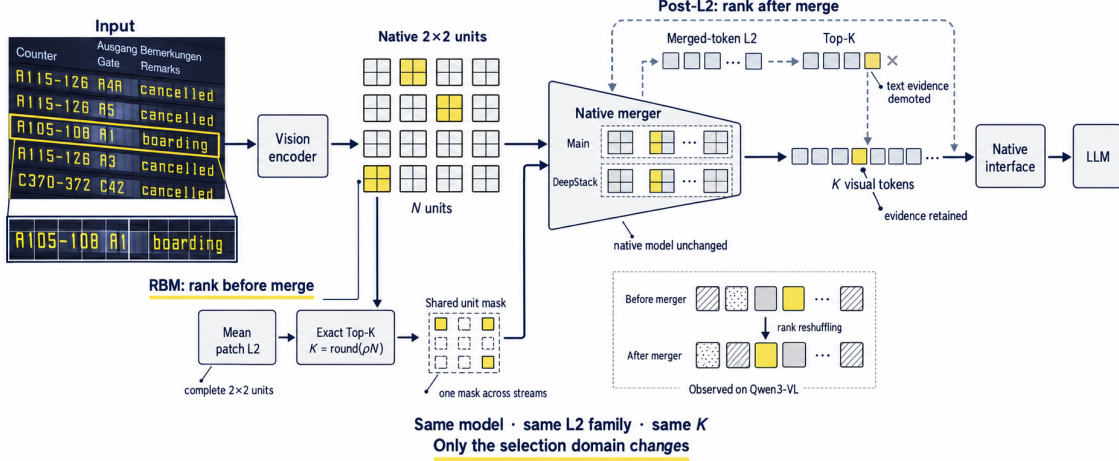


Figure 1: RBM follows the native merge topology. It ranks complete merger-input units, preserves a fixed per-image budget, and passes the selected units through the unmodified merger and language-model interface. The comparison changes selection order, not model weights.

demotes text-stroke-rich units. A locked RankBridge probe also shows that pre-merger rank complements FastV attention.

We contribute: (1) an operator-order formulation identifying when selection commutes with a native merger; (2) RBM, a training-free compressor that changes only selection order; and (3) matched full-split results and controls that establish its accuracy, mechanism, efficiency, and workload boundary.

Related Work

Visual-token reduction may act inside the LLM through attention or progressive pruning [1, 2, 3, 4], or at the encoder output through [CLS] attention, graph propagation, fitted predictors, and multiple cues [5, 6, 7, 8, 9]. Unified systems combine saliency with diversity or text guidance [10, 11, 12].

Merging also changes representation. ToMe, VisionZip, and AdaptMerge use matching, dominant-token preservation, and language guidance [13, 14, 15]; Token-Packer learns a compact projector, while GlimpsePrune, VScan, and AnchorPrune adapt reduction to inputs or model families [16, 17, 18, 19]. These methods motivate treating a selector’s input representation as part of its compression rule.

Recent selectors move reduction earlier or vary it across tokens and layers [20, 21, 22, 23, 24]; optimize coverage, reconstruction, or sensitivity [25, 26, 27, 28, 29]; and model semantics, information flow, or head importance [30, 31, 32, 33]. Layerwise, object-centric, local/global, and hybrid designs preserve evidence in other ways [34, 35, 36, 37, 38, 39]. Document pruning and provenance audits expose OCR-specific failures [40, 41], while LMMs-Eval and EffiVLM-Bench support evaluation [42, 43].

RBM isolates the native-merger boundary. Qwen-VL concatenates four neighboring patches before LayerNorm and an MLP; InternVL3 uses pixel-shuffle downsampling [44, 45, 46]. RBM keeps these operators and their interface unchanged but

ranks their inputs, enabling a matched pre/post comparison at equal token rate.

Compression Before a Native Merger

When does selection commute with merging?

Let $U = \{u_i\}_{i=1}^N$ be the native merge units and let M be the model’s per-unit merger. A representation-dependent scorer produces $a_i = s_{\text{pre}}(u_i)$ before merging and $b_i = s_{\text{post}}(M(u_i))$ after merging. At budget k , the two decisions are

$$K_{\text{pre}} = \text{TopK}_k\{a_i\}, \quad K_{\text{post}} = \text{TopK}_k\{b_i\}. \quad (1)$$

With deterministic tie breaking, selection commutes with merging at budget k exactly when $K_{\text{pre}} = K_{\text{post}}$. A sufficient condition for agreement at every budget is $b_i = g(a_i)$ for a strictly increasing g . We measure the violation at budget k by

$$D_k = 1 - \frac{|K_{\text{pre}} \cap K_{\text{post}}|}{|K_{\text{pre}} \cup K_{\text{post}}|}. \quad (2)$$

For a fixed set, selection and a per-unit merger are compatible; disagreement arises when the merger changes the ranking that determines the set.

For compression analysis, define the visual-token rate and task distortion as

$$R = \frac{k}{N}, \quad \mathcal{D}(R) = 1 - \frac{S(R)}{S(1)}, \quad (3)$$

where $S(R)$ is the downstream benchmark score at rate R . A lower $\mathcal{D}$ means that more task-relevant information survives. RBM and Post-L2 have identical R , so their vertical separation isolates distortion introduced by the representation used for selection.

Rank-Before-Merge

For every image, RBM groups ViT patch features into the model’s native 2×2 units. For a unit u with input patch vectors $h_{u,1:4}$, it uses the query-blind control score

$$s_{\text{pre}}(u) = \frac{1}{4} \sum_{j=1}^4 \|h_{u,j}\|_2, \quad s_{\text{post}}(u) = \|M(h_{u,1:4})\|_2. \quad (4)$$

It retains $k = \max(1, \text{round}(\kappa N))$ units and applies the native merger. The contract selects complete units, enforces each image’s exact top- k budget, propagates their identities through main and deepstack streams, and leaves the merger, positions, interface, and LLM unchanged. Scoring costs $O(Nd)$ and heap selection $O(N \log k)$. RBM adds no parameters, question tokens, or LLM-attention pass.

Qwen’s merger acts independently per unit, so a fixed kept set produces the same merged representations; this is byte-identical on the verified Qwen3-VL path and separates ranking from forward-operator changes.

Kept visual units (blue) per selector — two OCR cases

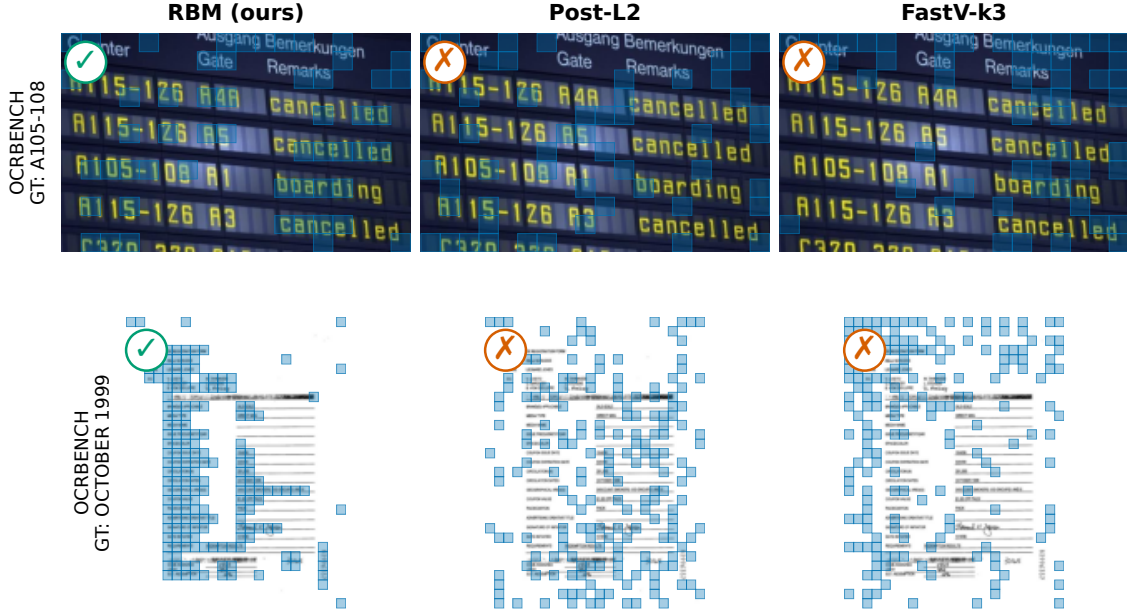


Figure 2: Audited OCRBench examples at 25% retention. RBM, Post-L2, and FastV-k3 preserve different spatial evidence. The examples illustrate selection behavior; aggregate results are reported separately.

A merger-aware prior for FastV

FastV ranks all visual tokens by question-conditioned attention at LLM layer $K = 3$. At $\kappa = 0.25$, RankBridge protects $q = \text{round}(\rho k)$ RBM-ranked positions and fills $k - q$ by FastV rank. A development gate over $\rho \in \{0.1, 0.2, 0.3\}$ selected and locked $\rho = 0.2$ for four $n = 200$ benchmarks. The endpoints are bit-identical to FastV ($\rho = 0$) and RBM in kept-set identity ($\rho = 1$); all candidates remain available until layer K .

Experimental Protocol

We evaluate Qwen3-VL-8B, Qwen2.5-VL-7B, and InternVL3-8B with bf16 weights, greedy decoding, and family-correct native positions. RBM, uncompressed, and Post-L2 use vLLM 0.19 V1 [47]; FastV/RankBridge use HuggingFace eager attention because FastV lacks a vLLM path. The eager native path matches 8/8 samples, and HF/vLLM uncompressed outputs agree on 16/16 held-out cases. Throughput is compared only within vLLM.

We use TextVQA VQA accuracy [48], DocVQA ANLS [49], OCRBench /1000 [50], and GQA normalized exact match [51]; full splits contain 5000, 5349, 1000, and 12578 items. Scorers port lmms-eval [42]. Pre/post tests pair samples and per-image token counts. Nine prespecified text-dense 25% comparisons use 20,000 paired-bootstrap resamples and sign-flip tests with Holm correction (all corrected $p \leq 4.5 \times 10^{-4}$). FastV/RankBridge are secondary paired probes on $n = 200$.

Table 1: Official full-split results. RBM and Post-L2 use the same per-image visual-token count. Δ is RBM minus Post-L2 at 25% retention. OCRBench uses points; other rows use percentage points.

Model	Benchmark (metric)	None	RBM	Post-L2	RBM-Post
Qwen3-VL	TextVQA (VQA acc.)	.844	.605	.222	+38.4
	DocVQA (ANLS)	.956	.481	.238	+24.3
	OCRBench (/1000)	760	547	184	+363
	GQA (EM)	.616	.449	.477	-2.8
Qwen2.5-VL	TextVQA (VQA acc.)	.862	.702	.442	+26.1
	DocVQA (ANLS)	.949	.636	.526	+11.0
	OCRBench (/1000)	817	480	182	+298
	GQA (EM)	.604	.559	.585	-2.6
InternVL3	TextVQA (VQA acc.)	.8338	.7890	.4148	+37.4
	DocVQA (ANLS)	.9221	.7284	.3820	+34.6
	OCRBench (/1000)	852	753	321	+432
	GQA (EM)	.6293	.5993	.6031	-0.4

Compression Results

Matched pre- and post-merger selection

Table 1 isolates the operator-order effect within query-blind L2 scoring. At 25% retention, RBM wins all nine text-dense cells by 11.0–38.4 pp or 235–432 OCRBench points across three VLMs and two merger designs. The gap widens at 12.5% on the reported text-dense cells, although RBM remains lossy relative to no compression.

GQA marks the boundary: Qwen rankings favor Post-L2 by 2.6–2.8 pp, while InternVL3’s -0.4 pp interval crosses zero. A full-split Qwen3-VL control matched at the merger input still gives RBM +27.68 pp TextVQA, +4.59 pp DocVQA, and +235 OCRBench, but -5.64 pp GQA (95% CI $[-6.42, -4.86]$). Selection stage is workload-conditioned.

FastV and the merger-aware prior

At the same 25% budget, RankBridge improves FastV-k3 on all four probes (Table 2). TextVQA gains 3.2 pp ($z = 2.11$); the other 0.5–1.6 pp gains are not significant at $n = 200$. RBM remains 11.6 pp stronger on OCRBench, so the hybrid misses its preset within-1 pp gate. Pre-merger rank adds information to query attention, but one global mixture cannot erase the workload boundary.

On full OCRBench, RBM exceeds FastV by 14.6 pp on Qwen3-VL and 8.7 pp on Qwen2.5-VL; FastV is stronger on Qwen3-VL TextVQA/GQA and Qwen2.5-VL TextVQA. The selectors therefore occupy different regimes.

Figure 3 gives rate-distortion operating points: pre-merger selection lowers text-dense distortion at equal rate, while GQA reverses the ordering.

Table 2: Locked Qwen3-VL probe at 25% retention ($n = 200$). RankBridge protects 20% of its final budget with the pre-merger rank and fills the remainder with FastV-k3 attention.

Benchmark	RBM	FastV-k3	RankBridge	RB-FastV (pp)
TextVQA	.5967	.7633	.7950	+3.2
DocVQA	.4239	.5863	.6023	+1.6
OCRBench	.5801	.4586	.4641	+0.6
GQA	.4150	.5050	.5100	+0.5

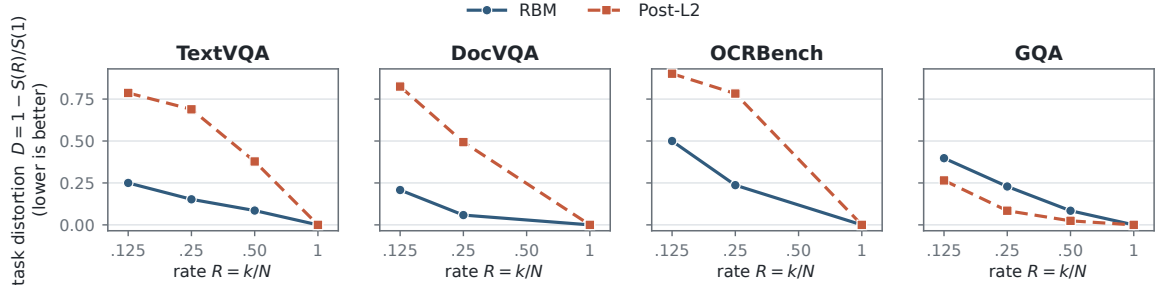


Figure 3: Task rate–distortion operating points on the locked Qwen3-VL diagnostic subset ($n = 200$ per point). Rate is the retained native visual-unit fraction; distortion is the relative loss in official task score from the uncompressed model. Curves connect measured rates only.

Why the Merger Changes Selection

Three Qwen3-VL diagnostics explain the operator-order gap (Fig. 4). Pre/post ranks correlate weakly: Spearman correlations are 0.137, 0.332, and 0.360, while top-25% Jaccard overlaps are 0.180, 0.243, and 0.278 on DocVQA, TextVQA, and GQA. The merger changes the top- k set, not only score scale.

The change is structured. Correlation between post-minus-pre rank shift and Sobel edge energy is +0.439/ + 0.155/ + 0.036. On DocVQA, RBM-only units have mean edge energy 0.641 versus 0.124 for Post-L2-only units; 92% versus 35% exceed the per-image median. This associates the merger with text-stroke demotion without attributing it to the training objective.

A ranking-swap control holds the post-merger forward path fixed but uses the pre-rank. It exactly selects RBM’s set (Jaccard = 1.000), recovers its accuracy, and yields byte-identical retained outputs. The Qwen3-VL gap therefore comes from ranking, not a different operation on kept units.

What the native merger changes

per-image distributions | 25% retained

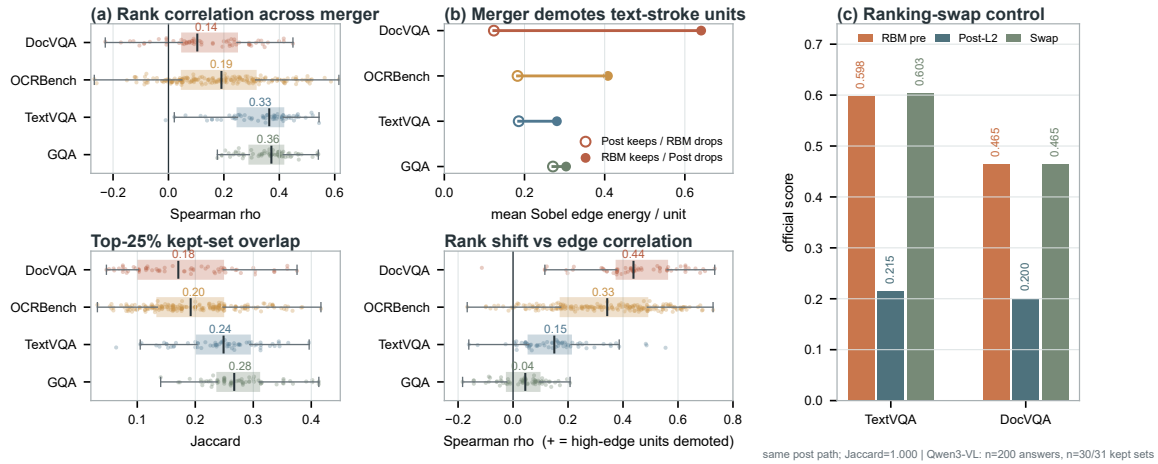


Figure 4: The native merger rewrites selection saliency. Rank agreement is low, text-stroke-rich units are demoted, and a ranking-swap control restores RBM accuracy with the same kept set.

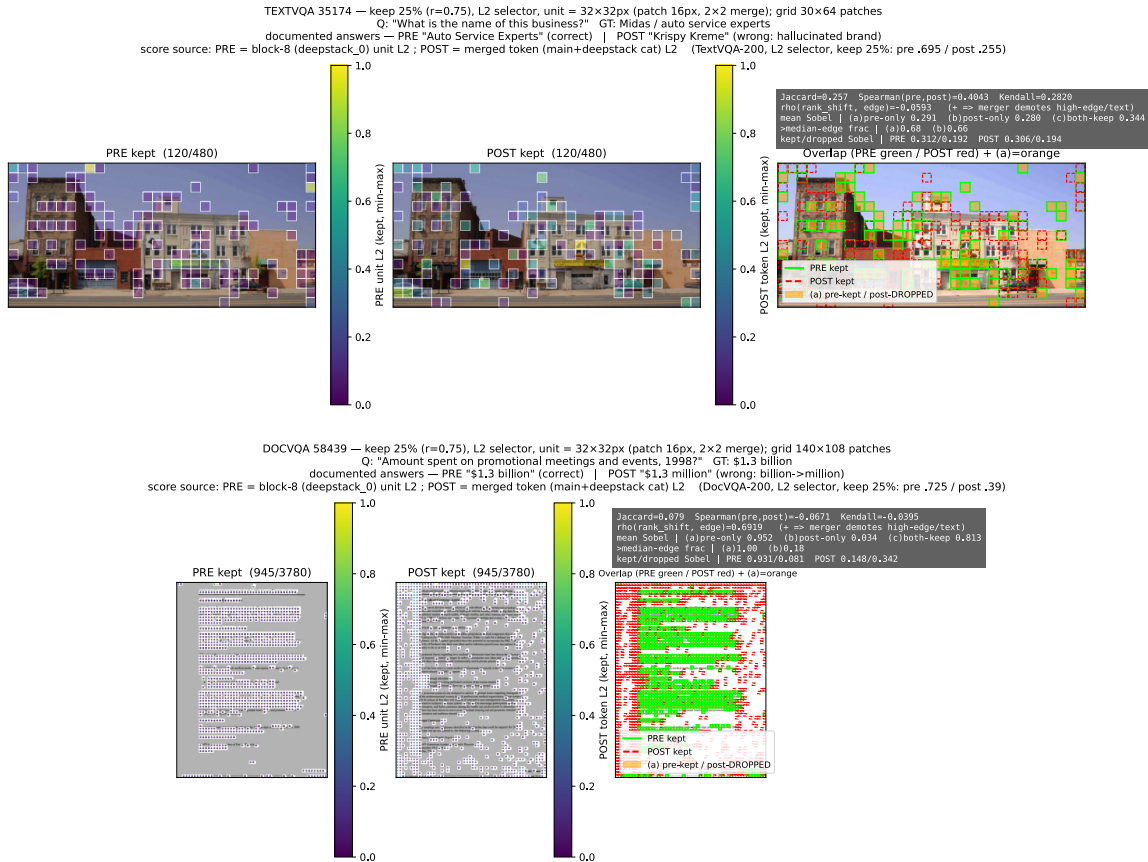


Figure 5: Token-survival maps for audited TextVQA and DocVQA images. Each row shows RBM, Post-L2, and their overlap at the same 25% retention. Green and red outlines identify units kept before and after merging, respectively. These cases illustrate spatial allocation rather than aggregate accuracy.

Efficiency and Scope

On one A40 with vLLM, RBM versus Post-L2 throughput is 4.61 versus 4.52 at 75%, 5.39 versus 5.54 at 50%, and 6.91 versus 6.89 requests/s at 25%; five repeats stay within 3%. RBM gains 12%, 31%, and 68% over no compression; at 25%, latency falls from 0.36 to 0.23 seconds. InternVL3 gains 1.8–2.5 $\times$. RBM leaves ViT work before its scoring tap intact and accelerates the merger and LLM sequences.

Byte-exact attribution holds for Qwen3-VL; Qwen2.5-VL retains an ordering confound, and InternVL3 provides accuracy replication plus a bounded swap. Models without native merge units are outside scope. FastV/RankBridge use the eager harness, several secondary cells use $n = 200$, and only the TextVQA hybrid gain is significant. Family pixel caps also preclude absolute cross-model comparison. The supported claim is merger-aware, text-dense compression, not a universal selector.

Conclusion

A native merger changes the ranking on which later compression depends. RBM therefore selects complete units before merging and leaves the surviving computation unchanged. At equal rates, it preserves substantially more text-dense evidence across three VLMs and two merger designs while accelerating inference. The GQA reversal bounds the claim: pre-merger rank is a strong text-rich signal, not a universal replacement for question-conditioned selection. Selection stage and native topology should be treated as part of VLM rate–distortion design.

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